Penlight

STUDY GUIDE

Fluids & Electrolytes: Potassium and Sodium Imbalances

Penlight sample lecture · 9 min lecture · October 17, 2026

Lite mode (offline) Sample content

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SECTIONS	KEY TERMS	FLASHCARDS	PRACTICE QS

Overview

This lecture (about 9 min, 1,436 words) covers Fluids Move, Potassium, Emergency Treatment of Severe Hyperkalemia and Potassium & T Wave. The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly. Always check the potassium level before you give any potassium and before you give digoxin.

What you should be able to do

- Explain the main ideas in “Fluids Move”.
- Explain the main ideas in “Potassium”.
- Explain the main ideas in “Emergency Treatment of Severe Hyperkalemia”.
- Explain the main ideas in “Potassium & T Wave”.
- Define active transport in your own words.
- Define diffusion in your own words.
- Define hyperkalemia in your own words.
- Recall the priority and safety points the instructor emphasized.

Must-know points

- The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly. 3:03
- The key point to remember is that peaked T waves are the earliest ECG sign of hyperkalemia. 7:10
- Notice that calcium, insulin, and albuterol only move or protect, but they do not remove the potassium from the body. 7:59
- Always check the potassium level before you give any potassium and before you give digoxin. 8:36
- Monitor intake and output, daily weight, and the neurological status for sodium problems. 8:58

Lecture notes

1 Fluids Move 0:00

- We will begin with how fluids move, then talk about sodium imbalances, and then spend the rest of our time on potassium, because potassium problems can stop the heart. 0:10
- Sodium is the major cation in the extracellular fluid. 0:43

- Hypotonic fluids, such as half-normal saline, have a lower concentration than plasma, so water moves into the cells. 1:53

> The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly. 3:03

- Treatment is to give fluids, usually hypotonic fluids, and to replace the water slowly. 3:30

potassium | sodium | intracellular fluid | active transport | hypernatremia | diarrhea

2 Potassium 3:40

- Hypokalemia is defined as a serum potassium level below three point five. 4:13
- Hyperkalemia is defined as a serum potassium level above five point zero. 6:16
- The causes include kidney failure, because the kidneys cannot excrete potassium, potassium-sparing diuretics such as spironolactone, ACE inhibitors, excessive potassium intake or supplements, tissue injury such as burns and crush injuries, and acidosis, which pushes potassium out of the cells. 6:22
- The signs of hyperkalemia also affect nerves, muscles, and the heart. 6:43

> The key point to remember is that peaked T waves are the earliest ECG sign of hyperkalemia. 7:10

potassium | hypokalemia | hyperkalemia | kidney | loop diuretic | T wave

3 Emergency Treatment of Severe Hyperkalemia 7:21

> Place the patient on a cardiac monitor and obtain the ECG. 7:21

> Notice that calcium, insulin, and albuterol only move or protect, but they do not remove the potassium from the body. 7:59

potassium | sodium | hyperkalemia | kidney | loop diuretic | cardiac monitor

4 Potassium & T Wave 8:20

> Always check the potassium level before you give any potassium and before you give digoxin. 8:36

> Monitor intake and output, daily weight, and the neurological status for sodium problems. 8:58

potassium | sodium | hypokalemia | hyperkalemia | T wave | cardiac monitor

Glossary

Active transport	The movement of particles against a concentration gradient and it requires energy.

Calcium	Notice that calcium, insulin, and albuterol only move or protect, but they do not remove the potassium from the body.
Cardiac monitor	Place the patient on a cardiac monitor and obtain the ECG.
Daily weight	Monitor intake and output, daily weight, and the neurological status for sodium problems.
Diarrhea	The causes include vomiting, diarrhea, excessive sweating, diuretics, and syndrome of inappropriate antidiuretic hormone secretion, or SIADH, in which the body holds on to too much water and dilutes the sodium.
Diffusion	The movement of solutes from an area of higher concentration to an area of lower concentration.
Digoxin	Patients who take digoxin and a loop diuretic need special attention.
ECG	Cardiac effects include weak pulses, dysrhythmias, and ECG changes such as a flattened T wave, a prominent U wave, and ST depression.
Extracellular fluid	Sodium is the major cation in the extracellular fluid.
Hyperkalemia	A serum potassium level above five point zero.
Hypernatremia	A serum sodium level above one hundred forty-five.
Hypokalemia	A serum potassium level below three point five.
Hyponatremia	A serum sodium level below one hundred thirty-five.
Insulin	Notice that calcium, insulin, and albuterol only move or protect, but they do not remove the potassium from the body.
Interstitial fluid	The fluid between the cells.
Intracellular fluid	The fluid inside the cells, and it holds about two thirds of the body's water.
Kidney	The causes include kidney failure, because the kidneys cannot excrete potassium, potassium-sparing diuretics such as spironolactone, ACE inhibitors, excessive potassium intake or supplements, tissue injury such as burns and crush injuries, and acidosis, which pushes potassium out of the cells.
Loop diuretic	Remove the potassium from the body with loop diuretics, sodium polystyrene sulfonate or other binders, and in severe renal failure, dialysis.

Osmosis	The movement of water across a semipermeable membrane from an area of lower solute concentration to an area of higher solute concentration.
Peaked T waves	The key point to remember is that peaked T waves are the earliest ECG sign of hyperkalemia.
Potassium	Always check the potassium level before you give any potassium and before you give digoxin.
Sodium	Sodium is the major cation in the extracellular fluid.
T wave	In hypokalemia, the muscles are weak, the bowel is slow, and the T wave is flat.

Quick review checklist

- [ ] The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly.
- [ ] The key point to remember is that peaked T waves are the earliest ECG sign of hyperkalemia.
- [ ] Notice that calcium, insulin, and albuterol only move or protect, but they do not remove the potassium from the body.
- [ ] Always check the potassium level before you give any potassium and before you give digoxin.
- [ ] Monitor intake and output, daily weight, and the neurological status for sodium problems.
- [ ] We will begin with how fluids move, then talk about sodium imbalances, and then spend the rest of our time on potassium…
- [ ] Hypokalemia is defined as a serum potassium level below three point five.
- [ ] Place the patient on a cardiac monitor and obtain the ECG.

Flashcard preview

Active transport
The movement of particles against a concentration gradient and it requires energy.

Diffusion
The movement of solutes from an area of higher concentration to an area of lower concentration.

Hyperkalemia
A serum potassium level above five point zero.

Hypernatremia
A serum sodium level above one hundred forty-five.

Hypokalemia
A serum potassium level below three point five.

Hyponatremia
A serum sodium level below one hundred thirty-five.

Interstitial fluid
The fluid between the cells.

Intracellular fluid
The fluid inside the cells, and it holds about two thirds of the body's water.

Osmosis
The movement of water across a semipermeable membrane from an area of lower solute concentration to an area of higher solute concentration.

The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly.

Self-test

1. Which term is described here? “The fluid between the cells.”

A. Interstitial fluid
B. Hypokalemia
C. Peaked T waves
D. Potassium

2. Which term is described here? “A serum potassium level below three point five.”

A. Hypernatremia
B. Intracellular fluid
C. Peaked T waves
D. Hypokalemia

3. Which term is described here? “The movement of solutes from an area of higher concentration to an area of lower concentration.”

A. Hypokalemia
B. Daily weight
C. Diffusion
D. Active transport

4. Which term is described here? “A serum sodium level above one hundred forty-five.”

A. Active transport
B. Hypotonic
C. Hypernatremia
D. Dehydration

5. Which term is described here? “A serum potassium level above five point zero.”

A. Diffusion

B. SIADH

C. Hypernatremia

D. Hyperkalemia

6. Which term is described here? “A serum sodium level below one hundred thirty-five.”

A. Dehydration

B. Hyperkalemia

C. Active transport

D. Hyponatremia

7. Which term is described here? “The movement of particles against a concentration gradient and it requires energy.”

A. Hyperkalemia

B. Calcium

C. Active transport

D. Intracellular fluid

8. Which term best completes this statement from the lecture? “The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give _____ replacement slowly.”

A. Hyponatremia

B. Sodium

C. Intracellular fluid

D. Hypokalemia

Answer key

1. A — The lecture defines Interstitial fluid: “The fluid between the cells.” (≈0:32)

2. D — The lecture defines Hypokalemia: “A serum potassium level below three point five.” (≈4:13)

3. C — The lecture defines Diffusion: “The movement of solutes from an area of higher concentration to an area of lower concentration.” (≈1:10)

4. C — The lecture defines Hypernatremia: “A serum sodium level above one hundred forty-five.” (≈3:13)

5. D — The lecture defines Hyperkalemia: “A serum potassium level above five point zero.” (≈6:16)

6. D — The lecture defines Hyponatremia: “A serum sodium level below one hundred thirty-five.” (≈2:36)

7. C — The lecture defines Active transport: “The movement of particles against a concentration gradient and it requires energy.” (≈1:20)

8. B — The lecture states: “The nurse should monitor neurological status, restrict fluids if the cause is SIADH, and give sodium replacement slowly.” (≈3:03)

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